Independent review: C0 Cantor reduction

1 November 2026

Fresh isolated replay passed seven modules and 25 axiom readbacks: twelve main declarations, six producer controls and seven independent controls. The checked polynomial correspondence holds at its stated scope. The written divisor-class derivation is consistent under its smooth-curve assumptions and is separate from machine verification.

Native polynomial correspondence

Native `three_leads.rs:1015-1024` has norm $= -B$ and slope $= C$. Thus $C = 0$ reaches the general two-Bezout fallback. With $U = u_1u_2$, $v_i = \ell + au_i$, $f - \ell^2 = UQ$, and both Bezout identities, the literal numerator equals

$$\ell + U\{c_1a(e_1 + e_2) + c_2(a^2 + Q)\}.$$

The proof permits arbitrary c_2; an independent control uses $c_2 = 1$. The initial remainder is ℓ by its degree bound. The exact monic quotient is $X - x$, and the output remainder is $-\ell(x)$. This describes one polynomial degree-4 to degree-1 reduction, conditional on native egcd/division specifications; it is not their execution proof.

Written geometric fidelity

On the smooth projective model of a squarefree monic quintic in characteristic not two, the ideals $(u_i, y - v_i)$ have quotient lengths two. Coprime supports separate the inputs. Away from ramification, $y + \ell$ is a local unit on the selected branch, so the multiplicities of $y - \ell$ equal those of $f - \ell^2$. At ramification, squarefreeness gives a simple root; local orders are $\operatorname{ord}(y) = 1$ and $\operatorname{ord}(x - \alpha) = 2$, so $y - \ell$ has order one. Repeated nonramified roots and residual/input overlap are retained with multiplicity.

At infinity, y has pole order five and ℓ has pole order at most four. Hence

$$\operatorname{div}(y - \ell) = D_1 + D_2 + R - 5O, \qquad \operatorname{div}(x_{\mathrm{coord}} - x) = R + R' - 2O.$$

The second equality includes ramified $R = R'$ with multiplicity two. In the actual quotient $\mathrm{Div}^0/\mathrm{Prin}$ these imply the output class $[R' - O]$, where $R' = (x, -\ell(x))$. No desired class-sum rule is assumed. The smooth curve, valuations and Picard carrier remain written mathematical inputs, not Lean constructions; original-field descent remains explicit.

Controls and provenance

Independent controls cover nonzero c_2, wrong quotient, ramification factorization/simple coefficient, ordinate sign and repeated residual support. They are algebraic examples, not formal local-valuation proofs. All readbacks use only the standard permitted axioms.

Published input commit: `ceecc0b38f9a3e14370781b2275afc6e59b27894`. An initial archive-suffix error produced empty modules; that invalid attempt is preserved and superseded by nonempty, byte-compared extraction and a 25-declaration audit. An ambiguous control identifier was also corrected with its failure retained. Review was source-first, not blind. Cached dependencies remain trusted. No scientific execution or new costs were produced.